

Experiment No.6

Aim: To study and implement Security as a Service on AWS/Azure.

Objectives:

- Develop a theoretical understanding of Security as a Service (SECaaS) and its role in enhancing the security posture of cloud environments.
- Apply practical knowledge to configure and utilize AWS security services.

Outcomes:

- demonstrate proficiency in utilizing AWS security services to enhance the security and compliance of their cloud-based applications and resources.
- gain applied knowledge in implementing Security as a Service on AWS.

Software Requirements: Windows/Ubuntu operating system, AWS Account

Theory:

Security as a Service (SECaaS) refers to the provision of security services and features in a cloud environment, allowing users to enhance the security of their applications and data without the need for on- premises hardware or complex infrastructure management. AWS provides a comprehensive set of security services, including Identity and Access Management (IAM) for managing user access, Key Management Service (KMS) for secure key storage and encryption, and AWS Web Application Firewall (WAF) for protecting web applications from common web exploits.

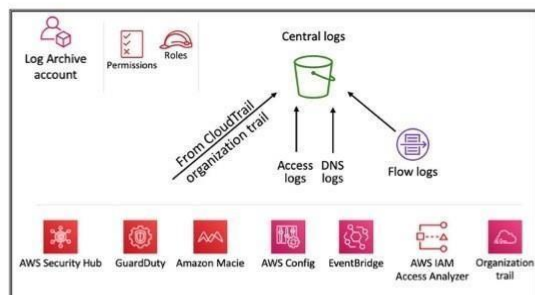
Key Components:

- **Identity and Access Management (IAM):** Granular control over who can access your AWS resources and what actions they can perform.
- **Key Management Service (KMS):** Securely generate, store, and manage encryption keys for your data.
- **Security Hub:** Centralized view of your security posture, aggregating security findings from various services.
- **Inspector:** Scans your Amazon Machine Images (AMIs) and EC2 instances for vulnerabilities.
- **WAF (Web Application Firewall):** Protects your web applications from common attacks.
- **Shield:** Distributed Denial-of-Service (DDoS) protection for your applications and resources.
- **CloudTrail:** Logs all API calls made to your AWS account, enabling audit and compliance.

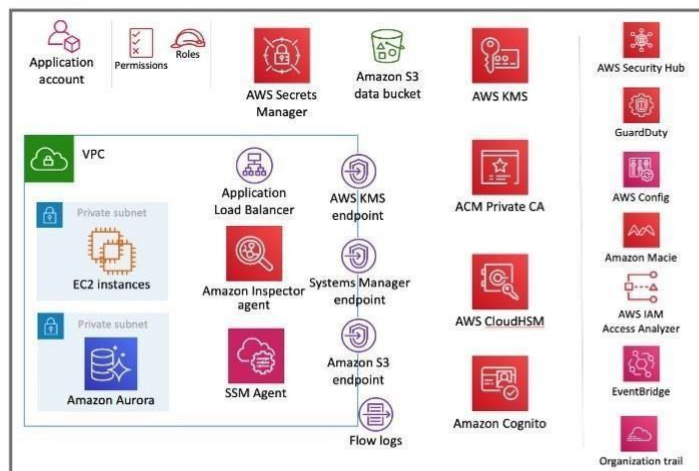
Organization



OU – Security



OU – Workloads



OU – Infrastructure

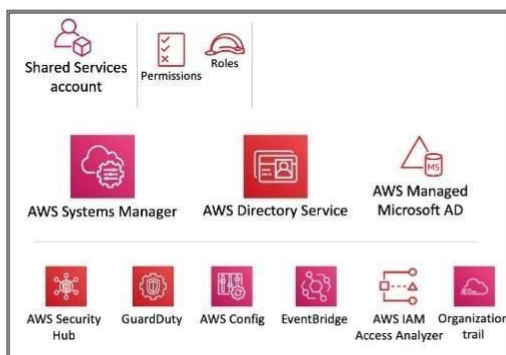
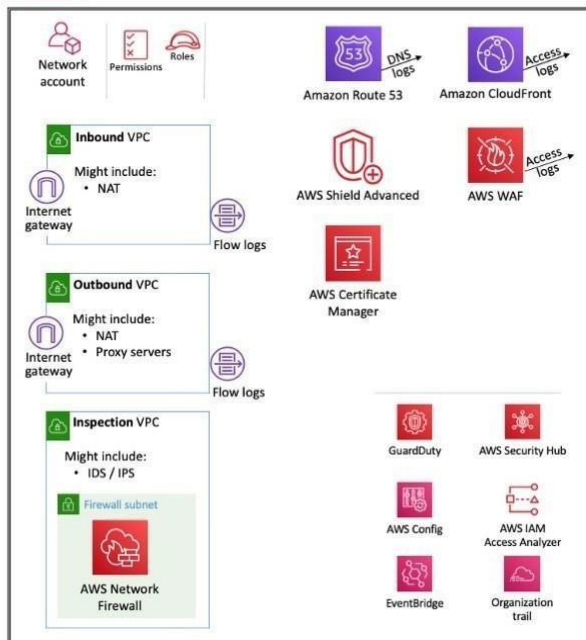
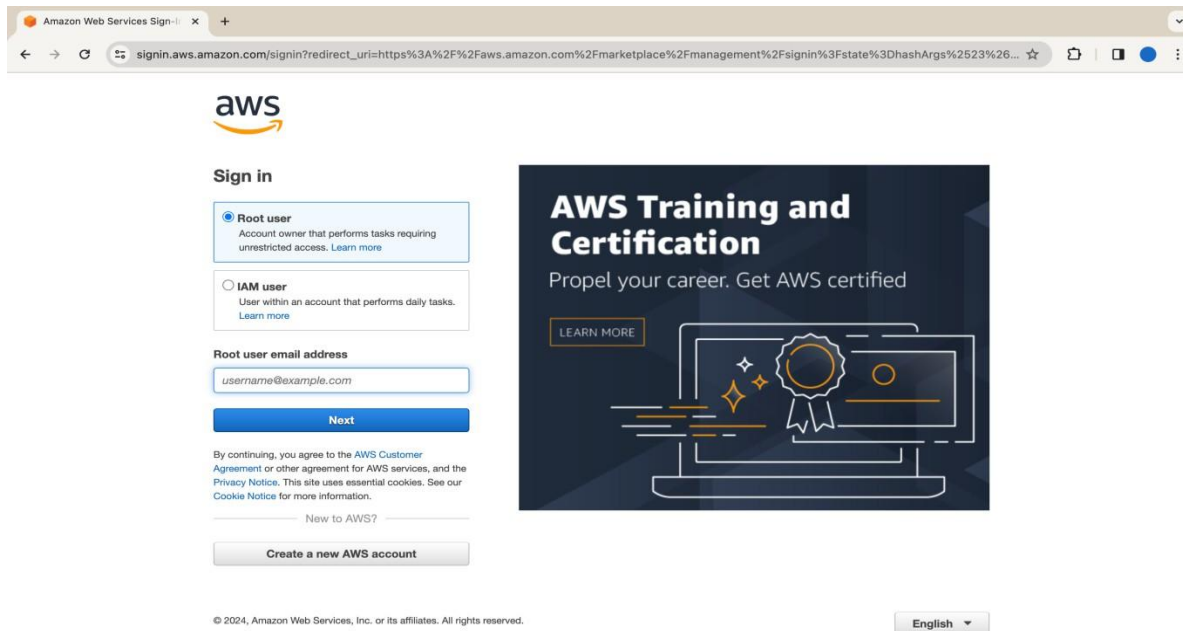


Figure 7 The AWS Security Reference Architecture

Stepwise Procedure to Implement Security as a Service using AWS:

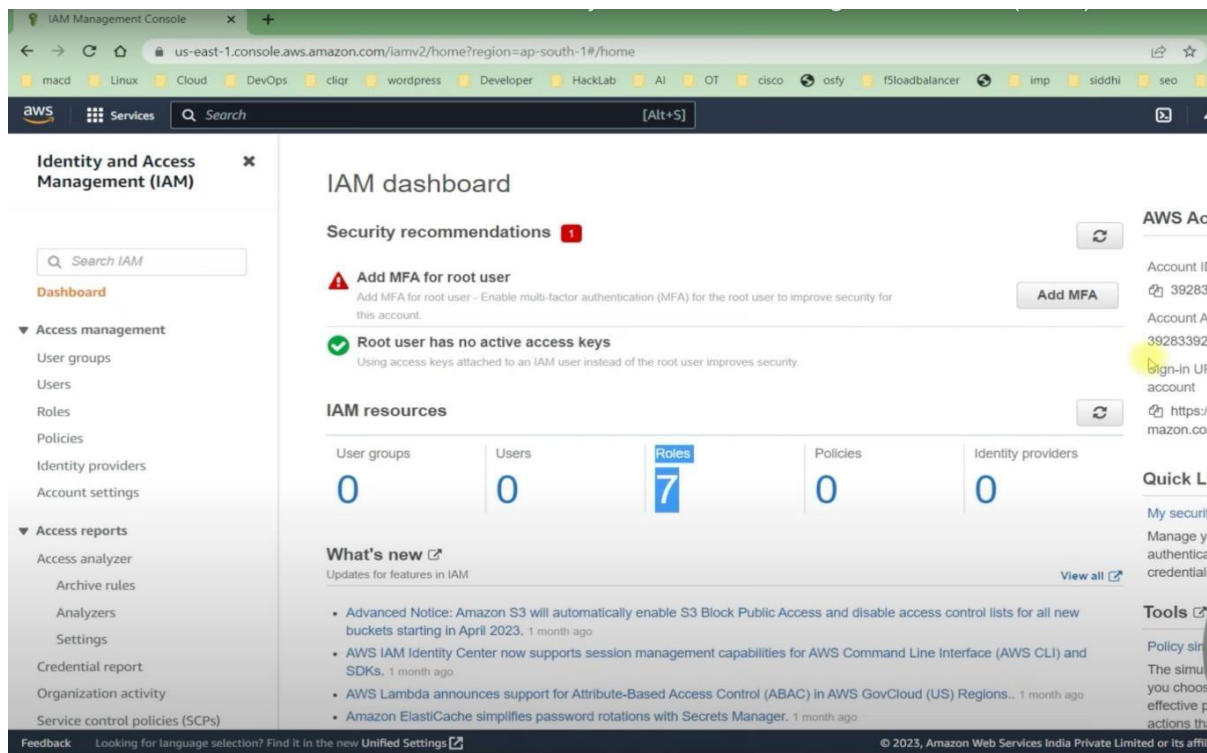
Step 1: Sign in to AWS Console

- Log in to the AWS Management Console using your AWS account credentials



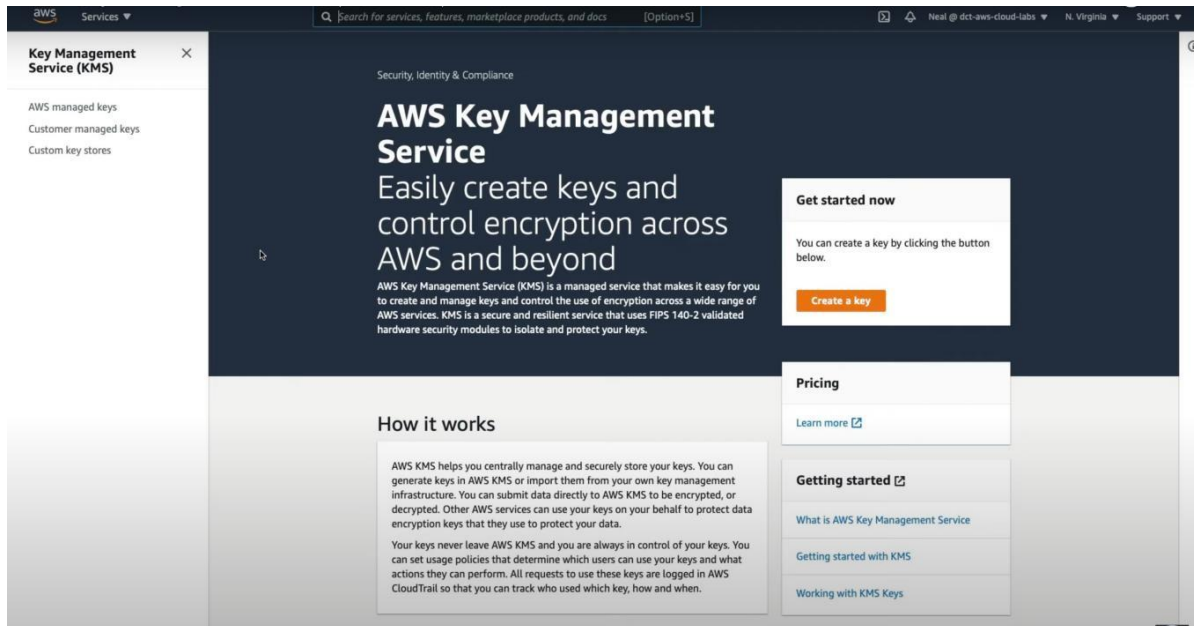
Step 2: Configure IAM for Access Management

- Navigate to IAM service in the AWS Console.
- Create IAM users, groups, and roles with appropriate permissions.
- Implement multi-factor authentication (MFA) for additional security



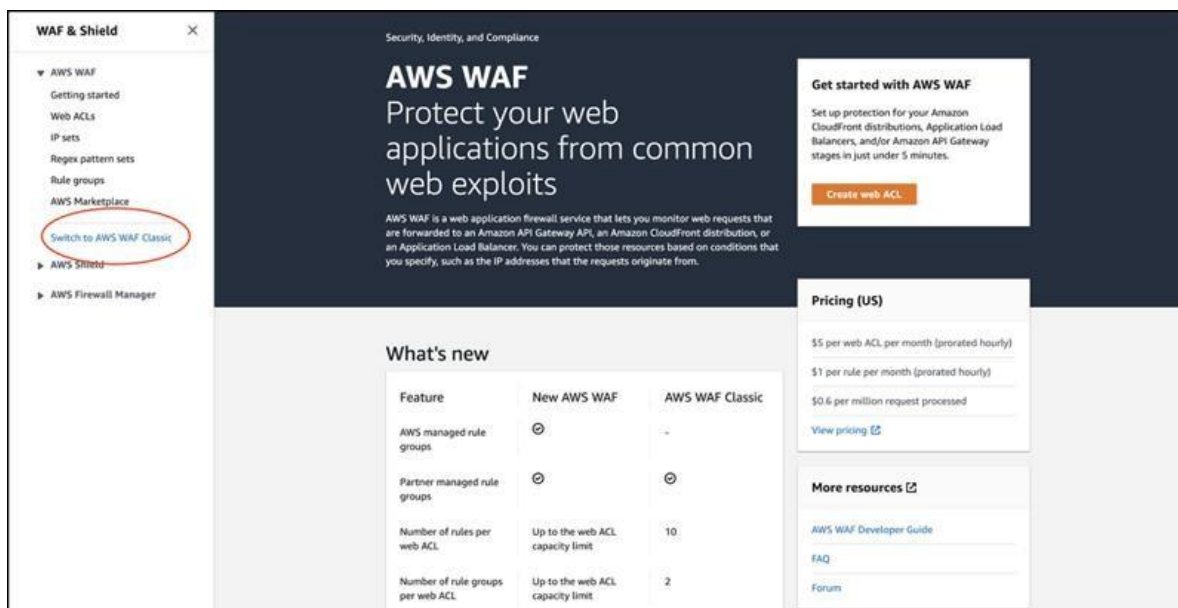
Step 3: Utilize KMS for Key Management

- Access the KMS service in the AWS Console.
- Create and manage cryptographic keys for securing sensitive data.
- Implement encryption for Amazon S3 buckets or EBS volumes using KMS.



Step 4: Implement AWS WAF for Web Application Security

- Navigate to AWS WAF service in the AWS Console.
- Create a WebACL (Web Access Control List) to define rules for web application protection.
- Configure rules to filter and block malicious traffic.



Conclusion: Thus, we have gained valuable insights into the implementation of Security as a Service on AWS.